

Arden Monaghan

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EDUCATION

University of Alberta

Sep 2022 – Dec 2026

Bachelor of Science in Computer Science, Specialization

Edmonton, AB

- Cumulative GPA: **3.6** — Major GPA: **3.8**
- **Relevant Coursework:** Data Structures and Algorithms, Introduction to Software Development, Machine Learning 1, Databases & SQL, Statistics 1 & 2, Computer Organization and Architecture, C Programming.
- **Relevant Awards:** Deans Research Award 2025, Jason Lang Scholarship, Deans Honor Roll x2

RELEVANT EXPERIENCE

Research Assistant

Jan 2025 – Present

University of Alberta Electrical and Computer Engineering Faculty

Edmonton, AB

- Implemented a **Computer Vision pipeline in OpenCV** to calibrate multiple cameras and estimate **3D hand-joint positions** using ArUco Marker detection, camera-pose estimation, and frame transformations.
- Integrated **SAM technology** using **PyTorch** to correctly compare 3D hand meshes to ground truth 2D masks. Fine-tuned model on dataset of hand-object representations.

Front-End Developer

Nov 2024 – Present

LitLife Co.

Edmonton, AB

- Managed the transition from Firebase to Supabase backend. Mapped **NoSQL to SQL schema** and reintegrated Flutterflow widgets, ensuring **accurate data mapping** and **secure storage**.
- **Revised user authorization** using **row-level security** systems to reduced unauthorized access attempts and harmful injections by **30%**.

Secretary

Apr 2024 – Present

Undergraduate Artificial Intelligence Society

Edmonton, AB

- **Leading** partnership with the **UA Innovation Fund** to assist in due diligence on **AI-based startups**, evaluating their technological innovation, market potential, and investment viability.
- Guiding over **100 students** per semester in understanding the fundamentals of data cleaning and supervised learning through scikit-learn workshops.
- **Coordinating** guest speaker sessions **with professors and industry**, creating an environment where experts can share valuable insights for students pursuing AI.

PROJECTS

Moodify | *Android Studio, Java, Firebase, Git*

Jan 2025 – Mar 2025

- **Developed and Integrated** Mood History UI/UX and Firebase querying, integrating real-time Firebase queries to enable users to share new moods and retrieve their posts.
- **Debugged and optimized** queries for mood creation, user-following, and history features using **JUnit** and **Firebase Test Lab**, reducing invalid data display errors by 25%.

MentorMatch UAlberta | *Python, Quart, PyQt6, Docker, Ollama*

Apr 2024 – Sept 2024

- Automated the task of matching Mentors and Mentees using an **LLM-integrated algorithm** for the **Science Mentorship Program** at UAlberta, reducing time spent matching from **10 days to 10 minutes**.
- Delivered a PyQt GUI desktop app that integrated Llama3 embeddings with Mentor matching criteria. Enhanced overall program **operations by over 50%** while allowing **student data to remain private**.

NHL Positivity Index | *Python, HuggingFace, LoRA, Seaborn*

Oct 2023 – Jan 2024

- Conducted **Sentiment Analysis** techniques on NHL subreddit data using Praw API to determine the positive sentiment scores **on all 32 NHL teams**. Featured on hockey platforms media sites and data events.
- **Fine-tuned** cardiffnlp/twitter-roberta-base-sentiment-latest with **5000 manually labeled** hockey comments to create Chelberta Model. Achieved an **accuracy score of 81.2**, an improvement from the base model from 79.2.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, Dart

Technologies/Frameworks: SQL, MongoDB, React.js, Quart, Qt, Supabase, FlutterFlow, PyTorch, OpenCV

Developer Tools: Git, Docker, AWS, Azure